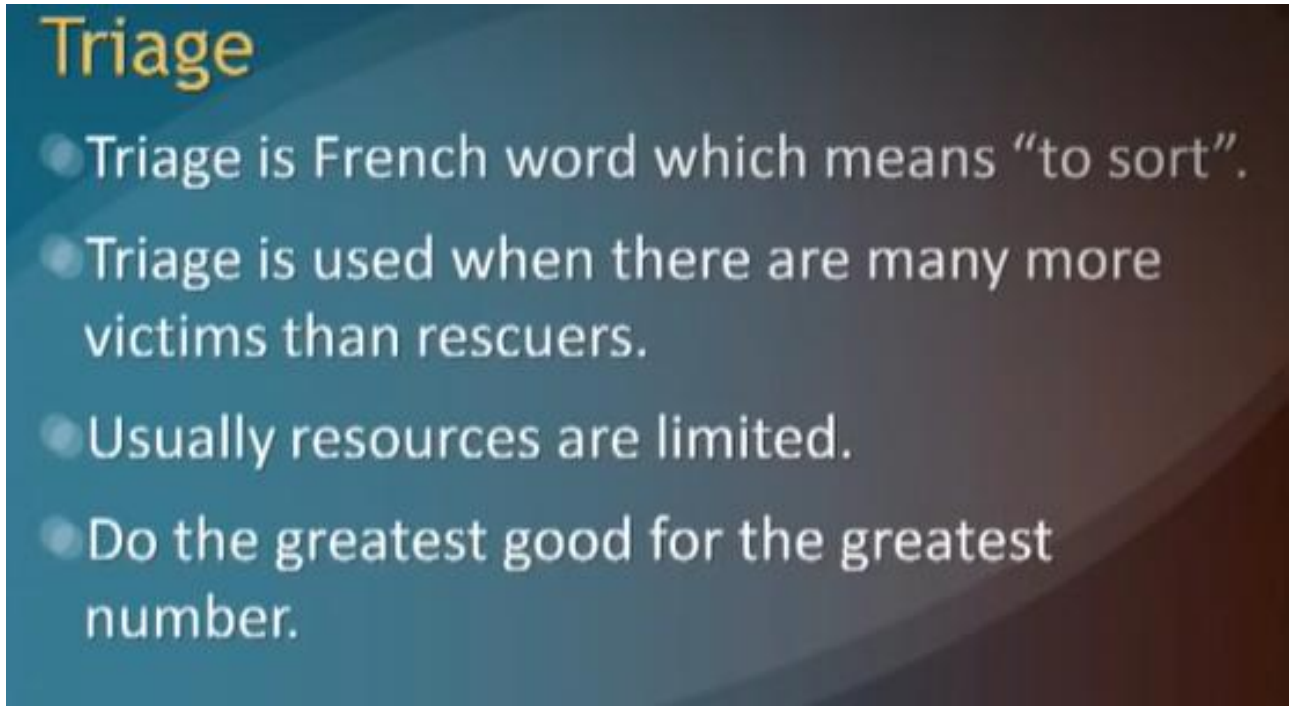
HealthCare Emergency and Triage

What is Triage

- The medical definition of triage is the **sorting and prioritization** of injured or sick people as well as disaster survivors according to their need for emergency care. Resource availability is also taken into account during this process.



https://www.verywellhealth.com/medical-triage-and-how-it-works-2615132
https://www.youtube.com/watch?v=9QHDs10e-G0

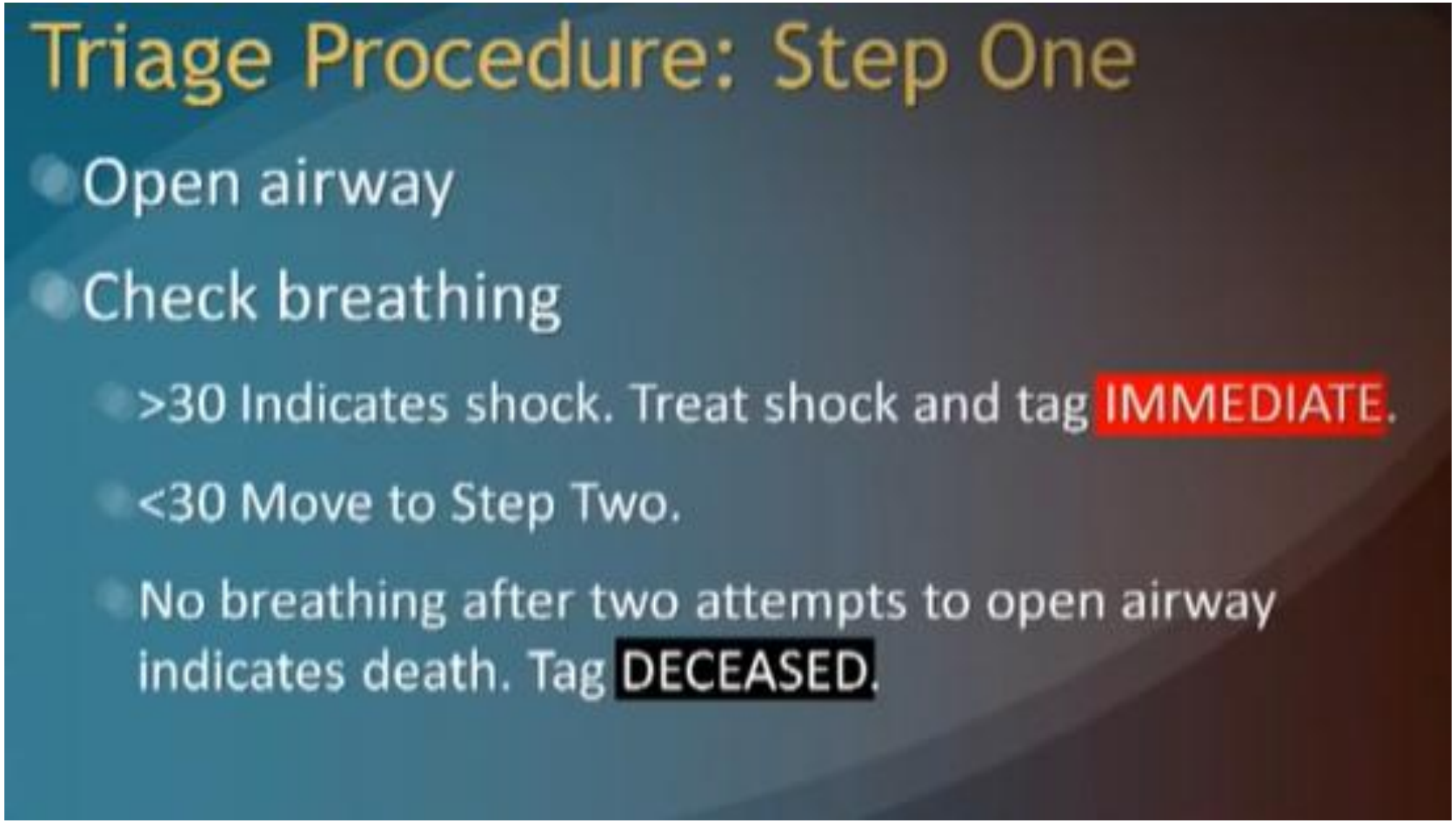
How Medical Triage Works

- Each organization has its own triage system. They all create priorities for who gets care or is transported for care. The most common triage systems use **color-coding.**
- Examples of the five categories or levels of triage that an organization may use:
- Red: The individual needs immediate attention for a critical life-threatening injury or illness and may be transported first for medical help.
- Yellow: The individual has serious injuries that need immediate attention. In some systems, yellow tags are transported first because they have a better chance of recovery than red-tagged patients.
- Green: This could indicate less serious or minor injuries that are non-life-threatening. Transport may be delayed.
- **Black:** This could indicate an individual is deceased or mortally wounded. It may mean that they are beyond help and, therefore, is a lower priority than those who can be helped.
- White: The individual doesn't have an injury or illness.



https://www.verywellhealth.com/medical-triage-and-how-it-works-2615132

Triage

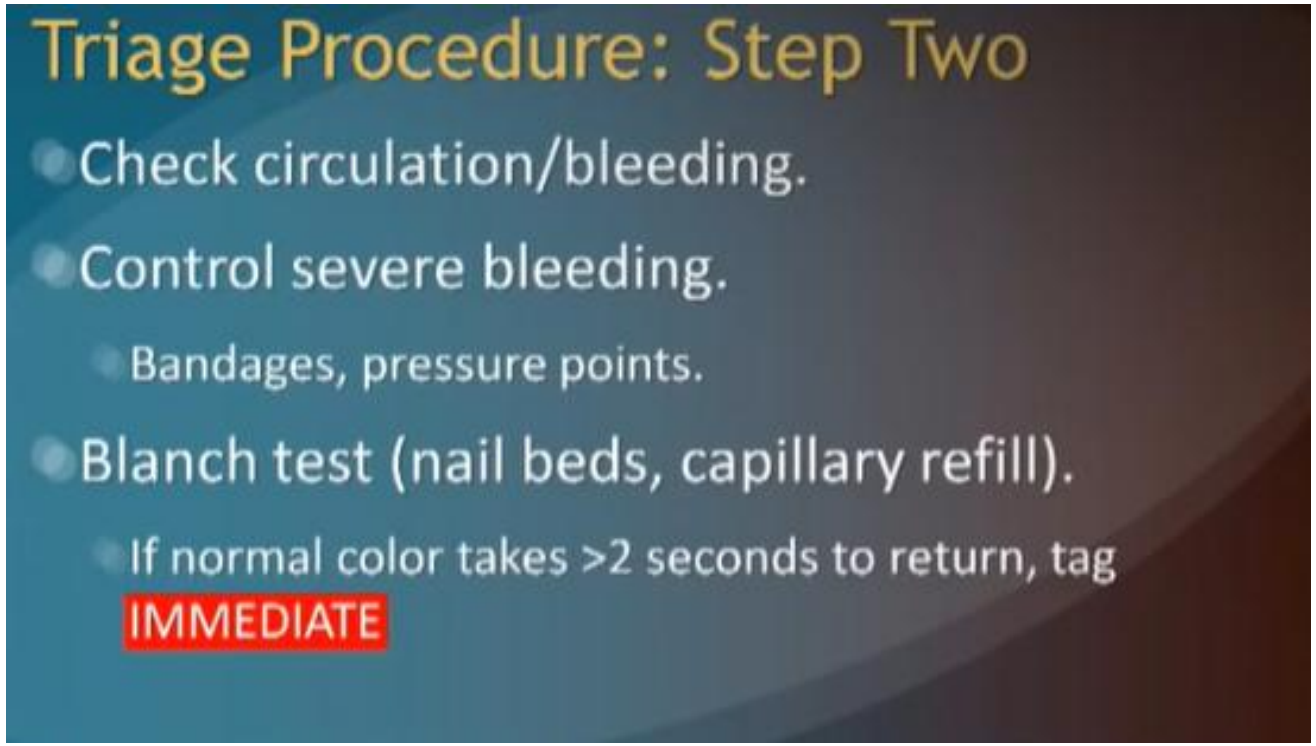


A normal respiratory rate for adults is typically between 12 to 20 breaths per minute. However, a significantly elevated or decreased respiratory rate can be a sign of various medical conditions, including shock.

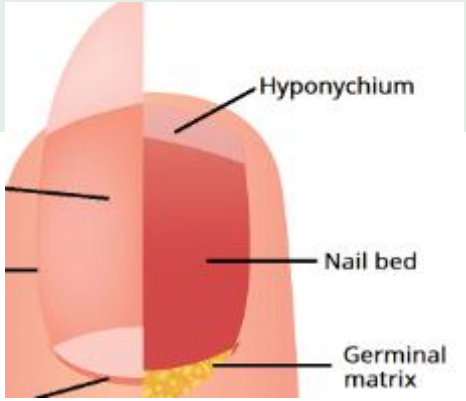
Shock is a critical condition that occurs when the body is not getting enough blood flow, which means the organs and tissues do not receive enough oxygen and nutrients to function properly. This can lead to serious damage or even death if not treated promptly.

https://www.youtube.com/watch?v=9QHDs10e-G0

Triage



The nail bed is the pinkish-colored soft tissue underneath your nail plate (the hard part of your nail).

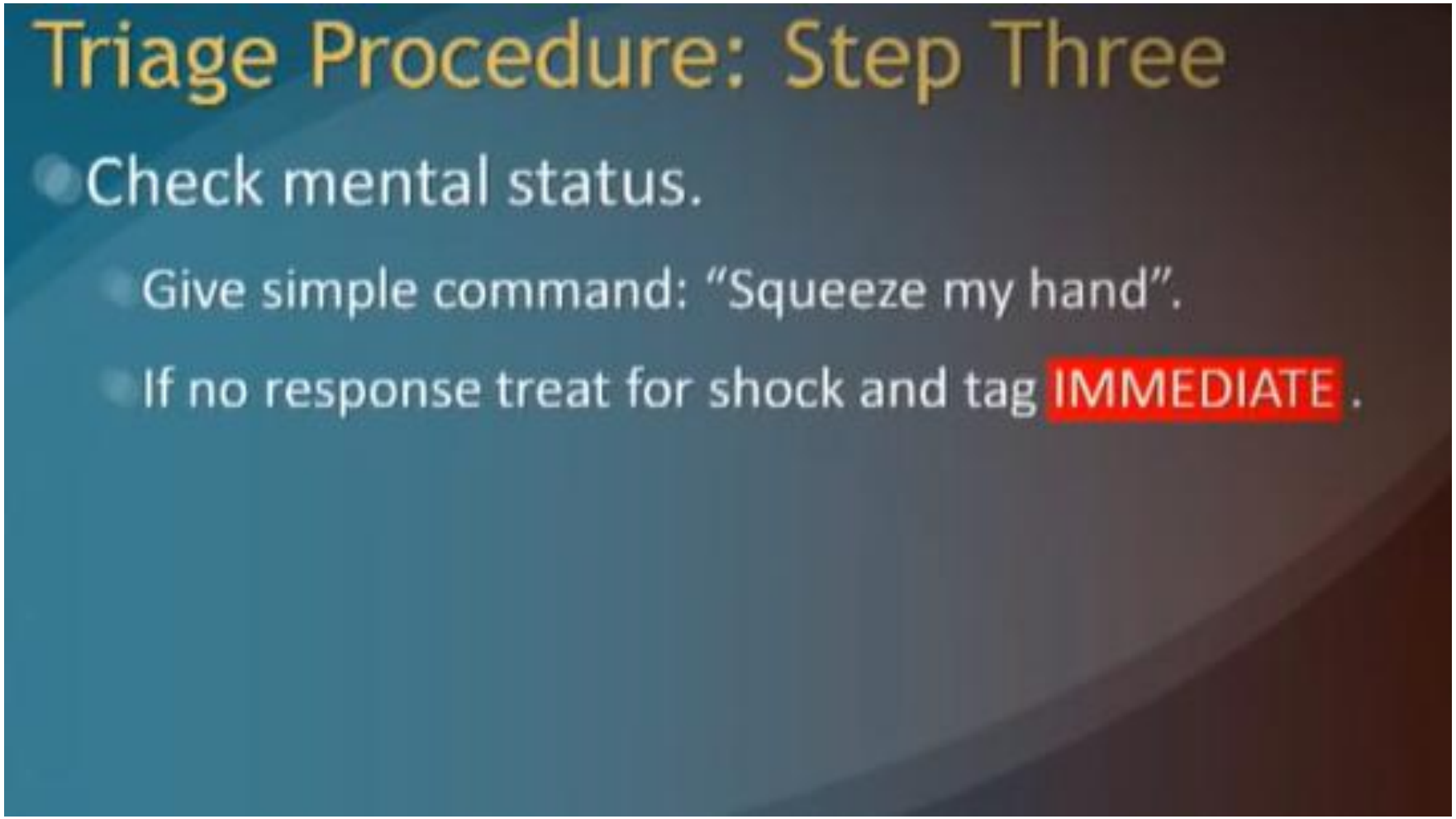


The blanch test, also known as the capillary refill test, is a simple procedure used to assess blood flow to the skin and underlying tissues. A healthcare provider presses down on the skin, usually on a fingertip or nail bed, until it turns white.

In a healthy individual, the color should return within 2 seconds. If it takes longer, it may indicate poor blood flow or other circulatory issues

https://www.youtube.com/watch?v=9QHDs10e-G0

Triage



https://www.youtube.com/watch?v=9QHDs10e-G0

Triage



https://www.youtube.com/watch?v=9QHDs10e-G0

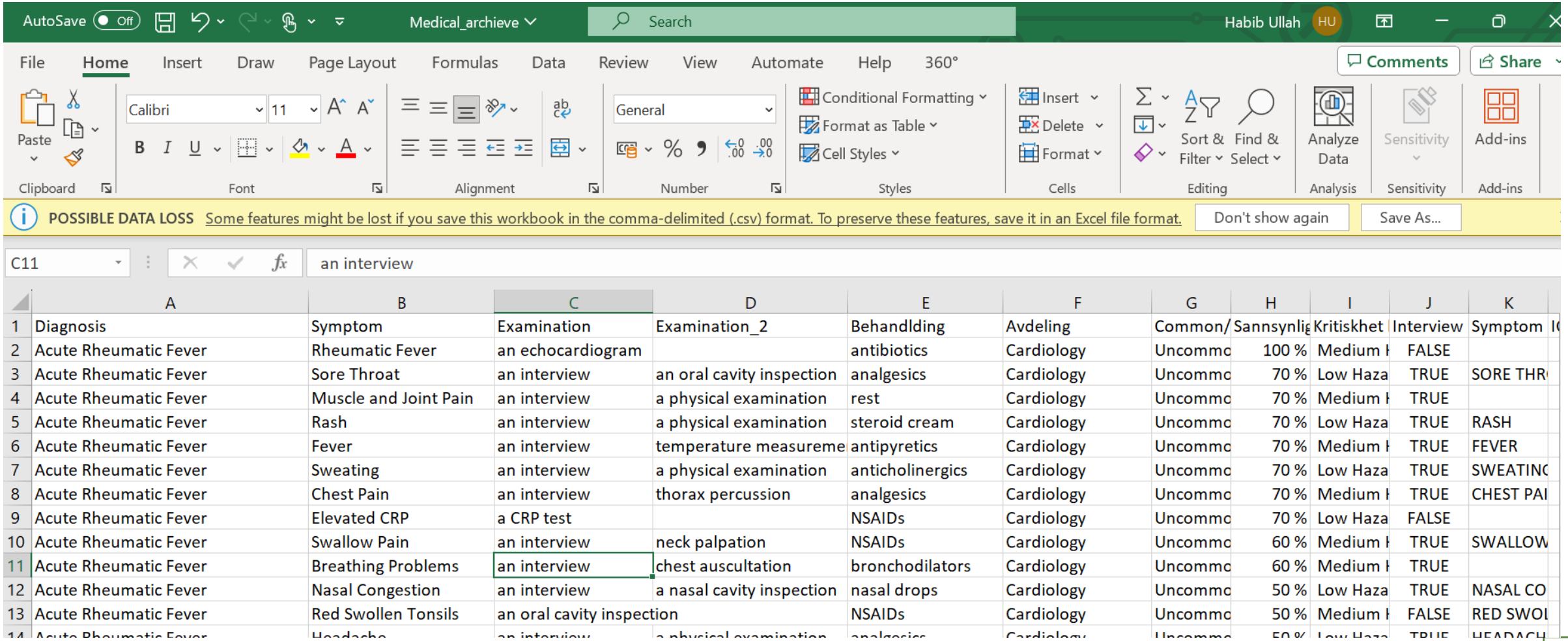
Course Projects related to Health

You will have access to two files to start with the course projects related to health:

Health_symptoms_patients_generation.ipynb.ipynb

Medical_Archieve.csv file

Medical_Archieve.csv file



	A	B	C	D	E	F	G	H	I	J	K
1	Diagnosis	Symptom	Examination	Examination_2	Behandlding	Avdeling	Common/	Sannsynlig	Kritiskhet	Interview	Symptom
2	Acute Rheumatic Fever	Rheumatic Fever	an echocardiogram		antibiotics	Cardiology	Uncommo	100 %	Medium H	FALSE	
3	Acute Rheumatic Fever	Sore Throat	an interview	an oral cavity inspection	analgesics	Cardiology	Uncommo	70 %	Low Haza	TRUE	SORE THR
4	Acute Rheumatic Fever	Muscle and Joint Pain	an interview	a physical examination	rest	Cardiology	Uncommo	70 %	Medium H	TRUE	
5	Acute Rheumatic Fever	Rash	an interview	a physical examination	steroid cream	Cardiology	Uncommo	70 %	Low Haza	TRUE	RASH
6	Acute Rheumatic Fever	Fever	an interview	temperature measureme	antipyretics	Cardiology	Uncommo	70 %	Medium H	TRUE	FEVER
7	Acute Rheumatic Fever	Sweating	an interview	a physical examination	anticholinergics	Cardiology	Uncommo	70 %	Low Haza	TRUE	SWEATING
8	Acute Rheumatic Fever	Chest Pain	an interview	thorax percussion	analgesics	Cardiology	Uncommo	70 %	Medium H	TRUE	CHEST PAI
9	Acute Rheumatic Fever	Elevated CRP	a CRP test		NSAIDs	Cardiology	Uncommo	70 %	Low Haza	FALSE	
10	Acute Rheumatic Fever	Swallow Pain	an interview	neck palpation	NSAIDs	Cardiology	Uncommo	60 %	Medium H	TRUE	SWALLOW
11	Acute Rheumatic Fever	Breathing Problems	an interview	chest auscultation	bronchodilators	Cardiology	Uncommo	60 %	Medium H	TRUE	
12	Acute Rheumatic Fever	Nasal Congestion	an interview	a nasal cavity inspection	nasal drops	Cardiology	Uncommo	50 %	Low Haza	TRUE	NASAL CO
13	Acute Rheumatic Fever	Red Swollen Tonsils	an oral cavity inspection		NSAIDs	Cardiology	Uncommo	50 %	Medium H	FALSE	RED SWOL

Medical_Archieve.csv file

There are in total 49 columns in the file.

```
'Diagnosis', 'Symptom', 'Examination', 'Examination_2', 'Behandlding',
'Avdeling', 'Common/uncommon', 'Sannsynlighet for kritisk utfall, når symptom er
påvist', 'Kritiskhet kategori', 'Interview reveal', 'Symptom.1', 'ICON',
'IsMainSymptom', 'DiscomfortLevel', 'PatientComplains', 'Hazard',
'PatientMobility', 'Examination 1', 'Examination 2', 'Examination 3',
'Treatment', 'ShameLevel', 'WalkAnim', 'WalkSpeedModifier', 'CollapseSymptom',
'RiskOfCollapseStartHours', 'RiskOfCollapseEndHours', 'CannotTalk',
'RiskOfDeathStartHours', 'RiskOfDeathEndHours', 'CollapseProcedure',
'PlanAllTreatments', 'BleedingLevel', 'Behandling (Engelsk)', 'Behandling
(Norsk)', 'Behandling beskrivelse (Engelsk)', 'Behandling beskrivelse (Norsk)',
'Krav', 'Surgery room', 'Diagnoses', 'Diagnoses_temp', 'Required Room(s)',
'Departement(s)', 'Required Equipment(s)', 'Legehåndbok', 'Virtuell sykehus',
'Kostnader($)', 'Sykdoms beskrivelse (Engelsk)', 'Sykdoms beskrivelse (Norsk)'
```

Medical_Archieve.csv file

The first column is ‘Diagnosis’

Acute Rheumatic Fever: an inflammatory illness that can occur as a result of an untreated or inadequately treated bacterial infection, particularly throat.

Angina Pectoris: chest pain or discomfort that occurs when the heart muscle (myocardium) does not receive an adequate supply of blood and, consequently, oxygen.

Aortic rupture: a severe and life-threatening medical condition in which the largest artery in the human body, tears or bursts open.

Atherosclerosis – Carotides: chronic condition characterized by the gradual buildup of fatty deposits, cholesterol, and other substances (plaque) on the inner walls of arteries, leading to the narrowing and hardening of the arteries.

Cardiac blunt injury: injury to the heart that occurs as a result of a blunt force impact to the chest. This type of injury can lead to damage to the heart muscle or other structures within the heart, and it is typically associated with high-impact accidents or incidents.

Medical_Archieve.csv file

The Second column is ‘Symptom’
An expected symptom a diagnosed disease should have. Column 13 is also ‘Symptom’. But they represent symptom which is indeed revealed by an interview.

Rheumatic Fever
Sore Throat
Muscle and Joint Pain
Rash
Fever
Sweating
Chest Pain
Elevated CRP (C-reactive protein, a substance produced by the liver due to inflammation or infection in the body.
Swallow Pain
Breathing Problems
Bradycardia
Arterial Stenosis
Chest Pain
Fatigue

Medical_Archieve.csv file

The Third column is ‘Examination’

an echocardiogram
an interview
a CRP test
an interview
an interview
an interview
an oral cavity inspection
an interview
heart monitoring
angiography
an interview
an interview
an interview

The fourth column is ‘Examination 2’

an oral cavity inspection
a physical examination
a physical examination
temperature measurement
a physical examination
thorax percussion

neck palpation
chest auscultation
a nasal cavity inspection

a physical examination
an ECG

thorax percussion

Medical_Archieve.csv file

The fifth column is ‘Behandlding’ (Treatment)

antibiotics
analgesics
rest
steroid cream
antipyretics
anticholinergics
analgesics
NSAIDs
NSAIDs
bronchodilators
nasal drops
NSAIDs
analgesics

The sixth column is ‘avdeling’ (department)

Cardiology
Emergency
internal medicine
general medicine
Neurology
Ortho

Medical_Archieve.csv file

The seventh column is ‘Common/uncommon’

Uncommon
Uncommon
Uncommon
Uncommon
Uncommon
Common
Common
Common
Common
Common

The eighth column is ‘Sannsynlighet for kritisk utfall, nÃ¥r symptom er pÃ¥vist’
Probability of critical outcome, when symptoms are detected

100 %
70 %
60 %
60 %
50 %
50 %
50 %
30 %
100 %

Medical_Archieve.csv file

The ninth column is ‘Kritiskhet kategori’
Criticality category

Medium Hazard
Low Hazard
Medium Hazard
Low Hazard
Medium Hazard
Low Hazard
Medium Hazard
Low Hazard

The tenth column is ‘Interview reveal’

FALSE
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
FALSE
TRUE

Medical_Archieve.csv file

The 11 column is ‘Kritiskhet kategori’
Criticality category

Medium Hazard
Low Hazard
Medium Hazard
Low Hazard
Medium Hazard
Low Hazard
Medium Hazard
Low Hazard

The 12 column is ‘Interview reveal’
The symptom in 13 colum is revealed in the interview or not.

FALSE
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
FALSE
TRUE

The 13 column is ‘Symptom’

SORE THROAT

RASH
FEVER
SWEATING
CHEST PAIN

SWALLOW PAIN

NASAL CONGESTION
RED SWOLLEN TONSILS
HEADACHE
BRADYCARDIA
ARTERIAL STENOSIS

Medical_Archieve.csv file

The 14 column is ‘ICON’	The 15 column is ‘IsMainSymptom’	The 16 column is ‘DiscomfortLevel’
0	FALSE	Low
		Medium
0	FALSE	Low
0	FALSE	Medium
0	FALSE	
0	FALSE	Medium
		Low
0	FALSE	Medium
0		
0	FALSE	None
0	FALSE	
0	FALSE	High
0	FALSE	None
0	TRUE	High
Fatigue.png	FALSE	High
	FALSE	

Medical_Archieve.csv file

The 17 column is ‘PatientComplains’	The 18 column is ‘Hazard’	The 19 column is ‘PatientMobility’
TRUE	Low	MOBILE
TRUE	Low	MOBILE
TRUE	Medium	MOBILE
TRUE	Low	MOBILE
TRUE	Medium	MOBILE
TRUE	Medium	IMMOBILE
TRUE	Low	
FALSE	Medium	
	Low	
	Medium	
	High	
	Medium	

Medical_Archieve.csv file

The 20 column is ‘Examination 1’	The 21 column is ‘Examination 2’	The 22 column is ‘Examination 3’
INTERVIEW	ORAL CAVITY INSPECTION	0
INTERVIEW	PHYSICAL AND VISUAL EXAMINATION	0
INTERVIEW	TEMPERATURE MEASUREMENT	0
INTERVIEW	PHYSICAL AND VISUAL EXAMINATION	0
INTERVIEW	THORAX PERCUSSION	0
INTERVIEW	NECK PALPATION	0
INTERVIEW	NASAL CAVITY INSPECTION	0
ORAL CAVITY INSPECTION	0	0
INTERVIEW	PHYSICAL AND VISUAL EXAMINATION	0
ECG	0	0
ANGIOGRAPHY	0	
	THORAX PERCUSSION	

Medical_Archieve.csv file

The 23 column is ‘Treatment’	The 24 column is ‘ShameLevel’	The 25 column is ‘WalkAnim’
	50	
ANALGESICS	50	0
	40	
STEROID TOPICAL CREAM	10	0
ANTIPYRETICS	40	0
ANTICHOLINERGICS	50	0
ANALGESICS	50	0
	0	
NSAIDS	50	0
	0	
NASAL DROPS	0	
NSAIDS	40	
	0	
	10	

Medical_Archieve.csv file

The 26 column is ‘WalkSpeedModifier’	The 27 column is ‘CollapseSymptom’	The 28 column is ‘RiskOfCollapseStartHours ’
0		
	0	
0		
0	0	4
0	0	
0	0	0,5
	0	0
0.5		1
	RESPIRATORY FAILURE	0
0		0,1
0	0	6
0	0	
0	SEPTIC SHOCK	6
0	0	0
0	HEART FAILURE	0
0	0	0

Medical_Archieve.csv file

The 29 column is ‘ RiskOfCollapseEndHours’	The 30 column is ‘CannotTalk’	The 31 column is ‘RiskOfDeathStartHours’
3	TRUE	0
0	0	2
0	0	0
	0	0
0	TRUE	0
0	0	
0		
0	0	
0	0	
24		
1		
0		
2		

Medical_Archieve.csv file

The 32 column is ‘ RiskOfDeathEndHours’	The 33 column is ‘CollapseProcedure’	The 34 column is ‘PlanAllTreatments’
0	0	0
4		
0	0	0
0	CONTROL PROCEDURE COLLAPSE	0
0	MEDIUM	0
0	0	0
	0	
	0	0
		All rows are 0

Medical_Archieve.csv file

The 35 column is ‘ BleedingLevel’	The 36 column is ‘Behandling (Engelsk)’	The 37 column is ‘Behandling (Norsk)’
	Antibiotics	Antibiotika
0	Analgesics	Analgetika
	Rest	**Hvile**
0	Steroid cream	Steroid krem
0	Antipyretics	Antipyretika
0	Anticholinergics	Antikolinergika
2	Analgesics	Analgetika
	NSAIDs	NSAIDs
0	NSAIDs	NSAIDs
	Bronchodilators	Bronkodilatatorer
0	Nasal drops	NesedrÃ¥per
5	NSAIDs	NSAIDs
	Analgesics	
	IV atropine	

Medical_Archieve.csv file

The 38 column is ‘ Behandling beskrivelse (Engelsk)’

Drugs in various forms used to treat various bacterial infections.

Drugs, mostly in pill form, used to relieve pain.

Prescription aimed at avoiding physical exertion in sick patients that would benefit

The 39 column is ‘Behandling beskrivelse (Norsk) ’

Legemidler i ulike former som brukes til Ã¥ behandle ulike bakterielle infeksjoner.
Legemidler, for det meste i pilleform, brukes til Ã¥ lindre smerte.
Resept rettet mot Ã¥ unngÃ¥ fysisk anstrengelse hos syke pasienter som ville ha godt av Ã¥ hvile.
En salve, lotion eller krem â€‹â€‹som inneholder kortikosteroider (stresshormon) som brukes i lokal behandling av ulike sykdommer.

The 40 column is ‘Krav’

Required room: Any office / Diagnostic unit
Required room: Any office / Diagnostic unit
Required room: Any office / Diagnostic unit
Required room: Any office / Diagnostic unit
Required room: Any office / Diagnostic unit
Required room: Any office / Diagnostic unit
Required room: Doctorâ€™s office / Observation room / ICU / Office / Diagnostic unit at Internal Medicine dpt / Cardiology dpt / Neurology dpt / DID / Traumatology dpt

Medical_Archieve.csv file

The 41 column is ‘ Surgery room’	The 42 column is ‘Diagnoses’	The 43 column is ‘Diagnoses_temp’
	Bacterial	0
FALSE	tonsillitis,Â Laryngitis,Â Bronchitis,Â Bacte	0
FALSE	rial gastroenteritis,Â Interstitial	Mitral vale stenosis
FALSE	nephritis,Â Pyelonephritis,Â Urinary tract	
FALSE	infection, most Internal med.	0
TRUE	diseases,Â Acute osteomyelitis,Â Acute	0
FALSE	rheumatic fever	
FALSE		0
FALSE	All diseases involving pain.	0
		0
	0	0
		0
	Athlete's foot,Â Dermatomyositis	

Medical_Archieve.csv file

The 44 column is ‘ Required Room(s)’	The 45 column is ‘Departement(s)’	The 46 column is ‘Required Equipment(s)‘
	Any	
Any office, Diagnostic unit	Emergency and Orthopeady	0
Any office, Diagnostic unit	Any	0
	Internal Medicine or Cardiology or	0
Office or Diagnostic unit	Neurology	0
Office or Diagnostic unit	Any	0
Office or Diagnostic unit	Emergency, Internal Medicine or	0
	Cardiology	0
Doctor's office or Observation	Any	0
room or ICU or	Any	0
OfficeorDiagnostic unit	General Surgery, Internal Medicine and	
	Cardiology	
	Cardiology	

Medical_Archieve.csv file

The 47 column is ‘ LegehÃ¥ndbok

https://nhi.no/sykdommer/hjertekar/ulike-sykdommer/giktfeber/
https://nhi.no/sykdommer/hjertekar/ulike-sykdommer/giktfeber/
https://nhi.no/sykdommer/hjertekar/ulike-sykdommer/giktfeber/
https://nhi.no/sykdommer/hjertekar/ulike-sykdommer/giktfeber/
https://nhi.no/sykdommer/hjertekar/ulike-sykdommer/giktfeber/

The 48 column is ‘Sykdoms beskrivelse (Norsk)’

Akutt revmatisk feber er en inflammatorisk sykdom som pÃ¥virker hjerte, muskler, ledd og hjerne. Vanligvis utvikler den seg etter en enkel streptokokkinfeksjon i halsen.
Akutt revmatisk feber er en inflammatorisk sykdom som pÃ¥virker hjerte, muskler, ledd og hjerne. Vanligvis utvikler den seg etter en enkel streptokokkinfeksjon i halsen.

The 49 column is ‘ ‘

Health



Course Project – Health01

Norwegian University of Life Sciences

Suggested questions:

Consider the diagnosis and Symptom related healthcare python script.

There are around 49 columns in the file.

Perform analysis considering the file to identify important columns/features for triage identification.

Are some columns/features irrelevant for triage patient identification ?

Which columns/features are most significant and why ?

Performn in depth analysis to combine some relevant or important columns/features for triage analysis.

Generate a random list of patients.

Use the identified features and identify the triage patients from the list based on the selected important columns/features.

Provide the context.

Applicability of tools ?

Six Sigma ?
Lean Six Sigma ?

Optimization techniques ?

Crisp DM ?

Course Project – Health02

Norwegian University of Life Sciences

Suggested questions:

Consider the diagnosis and Symptom related healthcare python script.

There are around 49 columns in the file.

Perform analysis considering the file to identify important columns/features for triage identification.

Are some columns/features irrelevant for triage patient identification ?

Which columns/features are most significant and why ?

Performn in depth analysis to combine some relevant or important columns/features for triage analysis.

Generate a random list of patients.

Could the identified features be used to train a machine learning model to predict the triage patients ?

You can use some classification or regression based machine learning models.

Provide context.

Applicability of tools ?

Six Sigma ?
Lean Six Sigma ?

Optimization techniques ?

Crisp DM ?

Hospital Simulator – HospitalSim for emergency - Triage

Hospical Simulator - RESOURCES

Multi-Agent Hospital Simulation with HospitalSim

An open-source, agent-based hospital management simulator — plus a ready-made emergency scenario — give you a realistic direction for studying triage, resource allocation, and operations under pressure.

01 **The Repository**

https://github.com/The-Swarm-Corporation/HospitalSim

02 **The Emergency Example**

https://github.com/The-Swarm-Corporation/HospitalSim/blob/main/examples/emergency_scenario.py

WHAT'S AHEAD

2 About HospitalSim

3 What's available

4 What it's for

5 The emergency scenario

6 What's in the script

7 Emergency project

WHAT IT'S ABOUT

The Repository: HospitalSim

github.com/The-Swarm-Corporation/HospitalSim

An open-source, MIT-licensed Python package from The Swarm Corporation that simulates a full hospital as a team of cooperating AI agents — executives, doctors, nurses, and reception staff — handling patients from check-in to discharge.

How the simulation flows

- Patient arrives → reception check-in → placed in a priority-scored queue
- Triage nurse assesses symptoms & vitals; emergency cases route to an ER doctor
- Doctor consults, using ChromaDB-backed patient history & similar-case matching
- Diagnosis, treatment, and EHR documentation are recorded, then patient is discharged
- In parallel, a CEO/CFO/CMO executive team reviews analytics for operational decisions

Quick Facts

PUBLISHER

The Swarm Corporation

LICENSE

MIT (free to use & modify)

INSTALL

pip3 install -U hospital-sim

BUILT ON

Swarms multi-agent framework

MEMORY LAYER

ChromaDB (RAG for patient records)

WHAT'S AVAILABLE

The Concrete Assets You Can Use

A Working Simulation Engine

- HospitalSimulation class — create a named hospital in one line
- generate_patients() and add_patient() for synthetic or custom cases
- run_simulation(duration_minutes, patient_arrival_rate) to drive the sim
- Agent roles: CEO/CFO/CMO, ER & general doctors, triage/floor nurses, receptionists
- Intelligent priority scoring & queue optimization out of the box
- Performance analytics: throughput, wait times, satisfaction, financials

Docs, Examples & Extras

- README with a full system-architecture flowchart (arrival → discharge)
- examples/ folder — runnable scripts, including emergency_scenario.py
- ehr_demo.py — a standalone look at the ChromaDB EHR / RAG system
- requirements.txt & pyproject.toml for a clean, reproducible setup
- MIT LICENSE file and a ready-made citation block for your report

PURPOSE

What HospitalSim Can Be Used For

1 Healthcare Operations Research

Model patient flow, staffing, and queueing to test how changes in resources or policy affect wait times and throughput.

2 Multi-Agent Systems Study

Study how independent AI agents (doctors, nurses, executives) coordinate, hand off tasks, and make decisions collaboratively.

3 Decision-Support Prototyping

Prototype and evaluate triage or resource-allocation algorithms in a safe, synthetic environment before ever touching real data.

4 AI + RAG Systems Education

See a working example of Retrieval-Augmented Generation (ChromaDB) applied to a realistic, structured records problem.

THE PROPOSAL

How to Turn This Into Your Project Proposal

1 Problem Statement

Pick an operational question: does priority-based queueing reduce wait times? How does staffing level affect patient throughput?

2 Data & Method

Install hospital-sim, generate or script synthetic patients, and run repeated simulations while varying one parameter at a time.

3 Scope & Objectives

Decide what you'll measure (wait times, throughput, cost) in emergency situation and what you'll change (arrival rate, staff mix, triage logic).

IN THE CONTEXT OF AN EMERGENCY

What emergency_scenario.py Is About

…/HospitalSim/blob/main/examples/emergency_scenario.py

A ready-made example script that stress-tests the simulator with three simultaneous, high-acuity patients — a possible heart attack, a possible stroke, and a possible ruptured appendix — arriving faster than in the basic demo. It shows the system handling urgent, life-threatening cases rather than routine visits.

Why this scenario matters

Realistic clinical complexity

Each patient has real symptom clusters, chronic conditions, current medications, and drug allergies — not toy data.

A true emergency-department stress test

patient_arrival_rate is raised to 0.2 (vs. 0.1 in the basic demo), simulating a faster, higher-pressure influx.

Built-in decision constraints

One patient is allergic to heparin, another to morphine — both common emergency drugs — forcing the system to reason around them.

IN THE CONTEXT OF AN EMERGENCY

Code Walkthrough: How the Script Is Built

```
from hospital_sim.main import (
    HospitalSimulation, Patient)

hospital = HospitalSimulation(
    "Emergency Medical Center")

emergency_patients = [
  Patient(
    name="Phoenix Razorcrest",
    age=45, gender="Male",
    chief_complaint="Chest pain
        radiating to arm",
    symptoms=["chest pain", ...],
    medical_history=["hypertension",
        "diabetes"],
    current_medications=[
        "metformin", "amlodipine"],
    allergies=[],
  ),
  # ...2 more Patient() entries
]

for patient in emergency_patients:
    hospital.add_patient(patient)

hospital.run_simulation(
    duration_minutes=20,
    patient_arrival_rate=0.2)
```

What's happening, step by step

1. Imports & setup

Pulls in the simulator and the Patient data class; a dedicated hospital instance is created just for this scenario.

2. Hand-built patients

Each Patient() call is fully authored by hand — not randomly generated — so every field (symptoms, history, meds, allergies) is deliberate.

3. Loading the queue

A simple for-loop feeds all three patients into the hospital via add_patient(), simulating simultaneous arrivals.

4. Running under pressure

run_simulation() is called with a shorter window (20 min) and a higher arrival_rate (0.2) than the basic demo — this is the 'emergency' lever.

IN THE CONTEXT OF AN EMERGENCY

The Three Patients, in Clinical Detail

Phoenix Razorcrest — 45, Male

Chest pain radiating to the arm

SYMPTOMS
Chest pain, left-arm pain, shortness of breath, sweating

HISTORY
Hypertension, diabetes

MEDICATIONS
Metformin, amlodipine

ALLERGIES
None reported

Classic presentation of a possible acute coronary event (heart attack)

Sage Whisperwind — 67, Female

Sudden severe headache

SYMPTOMS
Confusion, weakness, speech difficulty

HISTORY
Hypertension, atrial fibrillation

MEDICATIONS
Warfarin, metoprolol

ALLERGIES
Heparin

Sudden neuro symptoms + AFib/anticoagulant use suggest a possible stroke

Titan Shadowmere — 38, Male

Severe abdominal pain

SYMPTOMS
Nausea, vomiting, fever

HISTORY
Prior appendicitis

MEDICATIONS
None

ALLERGIES
Morphine

Fever + pain pattern + history raises concern for appendicitis recurrence

IN THE CONTEXT OF AN EMERGENCY

Related Project Directions to Explore

CLINICAL SAFETY

Allergy & Medication Conflict Audit

Verify — across many simulation runs — that the agents never recommend heparin for Sage or morphine for Titan. Log any near-misses.

OPERATIONS

Mass-Casualty Stress Test

Gradually raise patient_arrival_rate and patient count until wait times or care quality collapse; find the system's breaking point.

EVALUATION

Triage Accuracy Benchmark

Compare the simulation's priority ordering of the 3 patients against a real clinical standard like the Emergency Severity Index (ESI).

SCENARIO DESIGN

Custom Emergency Case Library

Author new high-acuity Patient() cases — pediatric trauma, overdose, anaphylaxis — and test how well the system generalizes.

COMPARATIVE STUDY

Emergency vs. Routine Comparison

Run the basic demo and emergency_scenario.py side by side; quantify how differently the system behaves under pressure.

AGENT DESIGN

New Specialist Agent

Extend the agent roster — e.g., add a cardiologist or radiologist agent — and see how it changes Phoenix's care pathway.

IN THE CONTEXT OF AN EMERGENCY · THE PROPOSAL

Proposed Project: ER Surge & Safety Evaluation

OBJECTIVE

Determine whether HospitalSim's agents can safely triage and treat multiple concurrent critical patients as arrival pressure increases — without missing a known allergy or medication conflict.

Background

- Built on the existing emergency_scenario.py (3 hand-authored critical patients)
- Two patients carry documented allergies (heparin, morphine) — common ER drugs
- Base case runs at duration_minutes=20, patient_arrival_rate=0.2

Method

- Re-run the base scenario 10–20 times to establish a performance baseline
- Sweep patient_arrival_rate upward (e.g., 0.2 → 0.5 → 1.0) and re-run at each level
- Add 2–3 new hand-authored patients with their own allergy/medication conflicts
- Log every agent decision: triage order, assigned doctor, treatment given

Evaluation Metrics

- Safety: % of runs with zero allergy/medication conflicts (target: 100%)
- Triage accuracy: agreement between simulated priority and an ESI-style ranking
- Operational: average wait time and throughput as arrival rate increases
- Breaking point: the arrival rate at which performance degrades sharply

Hospital simulation example:

Code

- In your first Colab cell, install the hospital-sim library

- !pip install -U hospital-sim

Code

- Because HospitalSim relies on multi-agent collaboration powered by AI (such as CEO, CFO, CMO, and doctor agents), you will need an active LLM API Key (such as an OpenAI API key) set in your environment.
- You can securely add your key to Google Colab by clicking the Secrets (key icon) in the left sidebar, adding a new secret named OPENAI_API_KEY, and pasting your API key.

Code

- import os
  from google.colab import userdata
- # Retrieve the API key from Colab's Secrets tab
  try:
- os.environ["OPENAI_API_KEY"] = userdata.get('OPENAI_API_KEY')
  print("API Key loaded successfully!")
  except Exception as e:
  print("API Key not found in Secrets. Please set it manually below:")
  os.environ["OPENAI_API_KEY"] = "your-actual-api-key-here"

Hospital simulation example:

Code

- from hospital_sim.main import HospitalSimulation, Patient
- # 1. Initialize the hospital simulation instance
  hospital = HospitalSimulation("Colab Emergency Medical Center")
-

```
# 2. Define the inputs (Emergency Patients)
emergency_patients = [
  Patient(
    name="Phoenix Razorcrest",
    age=45,
    gender="Male",
    chief_complaint="Chest pain radiating to arm",
    symptoms=[
      "chest pain",
      "left arm pain",
      "shortness of breath",
      "sweating",
    ],
    medical_history=["hypertension", "diabetes"],
    current_medications=["metformin", "amlodipine"],
    allergies=[],
  ),
  Patient(
    name="Sage Whisperwind",
    age=67,
    gender="Female",
    chief_complaint="Sudden severe headache",
    symptoms=[
      "sudden severe headache",
      "confusion",
      "weakness",
      "speech difficulty",
    ],
    medical_history=["hypertension", "atrial fibrillation"],
    current_medications=["warfarin", "metoprolol"],
    allergies=["heparin"],
  ),
  Patient(
    name="Titan Shadowmere",
    age=38,
    gender="Male",
    chief_complaint="Severe abdominal pain",
    symptoms=[
      "severe abdominal pain",
      "nausea",
      "vomiting",
      "fever",
    ],
    medical_history=["appendicitis"],
    current_medications=[],
    allergies=["morphine"],
  ),
]
```

IN THE CONTEXT OF AN EMERGENCY · THE PROPOSAL

Hospital simulation example:

```
# 3. Queue the patients in the hospital system
for patient in emergency_patients:
    hospital.add_patient(patient)

# 4. Execute the simulation
# duration_minutes: simulated length of time the simulation runs
# patient_arrival_rate: frequency of new random patients checking in
hospital.run_simulation(
    duration_minutes=20,
    patient_arrival_rate=0.2
)
```



Hospital simulation example:

```
2026-09-08 06:46:25 | INFO     | hospital_sim.main:add_patient:983 - Patient Sage Whisperwind checked in and added to queue
2026-09-08 06:46:25 | INFO     | hospital_sim.main:add_patient:959 - New patient arrival: Titan Shadowmere - Severe abdominal pain
```

Agent Name Crystal Moonwhisper [Loop: 1/1]

```
Hello, welcome. I'd be happy to help you get checked in.

Please share:
- Your full name
- Your age
- Your chief complaint, or the reason for today's visit
- When your symptoms started
- A phone number we can use to reach you
- Whether you have your ID and insurance card with you

If you're having severe symptoms or this is an emergency, please tell me right away.
```

```
2026-09-08 06:46:27 | ERROR    | swarms.agents.llm_manager:check_model_supports_utilities:341 - [Agent: Patient_Titan_Shadowmere] Model 'groq/moonshotai/kimi-
```

Agent Name Patient_Titan_Shadowmere [Loop: 1/1]

```
Hi, I'm Titan Shadowmere. I'm 38.

I'm here for severe abdominal pain. It started earlier today and it's gotten really bad. I've also been
nauseous, I've been vomiting, and I have a fever. This is for me, and it feels pretty urgent.

My phone number is 555-0187. I do have my ID and insurance card with me.
```

```
2026-09-08 06:46:29 | INFO     | hospital_sim.main:add_patient:983 - Patient Titan Shadowmere checked in and added to queue
✅ Patient Titan Shadowmere checked in and added to queue
   Reception interaction: Greet and check in the patient. Ask for their name, age, chief complaint, and gather initial informa...
   Patient response: You are at the hospital reception desk. The receptionist is checking you in and asking for informati...
📓 Starting Colab Emergency Medical Center simulation for 2 minutes
2026-09-08 06:46:29 | INFO     | hospital_sim.main:run_simulation:1662 - Starting Colab Emergency Medical Center simulation for 2 minutes (arrival rate: 0.2)
2026-09-08 06:46:29 | INFO     | hospital_sim.main:process_patient_queue:1017 - Processing patient from queue: Phoenix Razorcrest (Priority: 20)
2026-09-08 06:46:29 | INFO     | hospital_sim.main:process_patient_queue:1030 - Starting triage for Phoenix Razorcrest with Nurse Raven Stormborn
```

Hospital simulation example:

```
2026-09-08 06:46:29 | INFO     | hospital_sim.main:process_patient_queue:1030 - Starting triage for Phoenix Razorcrest with Nurse Raven Stormborn
                    Agent Name Nurse Raven Stormborn [Loop: 1/1]
Hello, I'm Nurse Raven Stormborn, one of the triage nurses in the emergency department. I'm going to ask you a
few questions so we can assess you properly and make sure you get the right care as quickly as possible.

What's your name, and what brought you to the hospital today?

2026-09-08 06:46:30 | ERROR    | swarms.agents.llm_manager:check_model_supports_utilities:341 - [Agent: Patient_Phoenix_Razorcrest] Model 'groq/dee

2026-09-08 06:46:30 | ERROR    | swarms.agents.llm_manager:check_model_supports_utilities:341 - [Agent: Patient_Phoenix_Razorcrest] Mode
                    Agent Name Patient_Phoenix_Razorcrest [Loop: 1/1]
Hi, I'm Phoenix Razorcrest.

I came in because I'm having chest pain, and it's going down my left arm. I'm also short of breath and sweating
a lot. I'm pretty scared, honestly.

                    Agent Name Nurse Raven Stormborn [Loop: 1/1]
Phoenix, thank you for telling me. I'm concerned about your symptoms, so I'm treating this as urgent right now.

I'm taking your vital signs now.

Your vital signs are:
- Blood pressure: 168/102
- Heart rate: 112
- Respiratory rate: 24
- Temperature: 98.6°F
- Oxygen saturation: 94% on room air

On a scale of 1 to 10, with 10 being the worst pain imaginable, what number is your chest pain right now?

2026-09-08 06:46:34 | ERROR    | swarms.agents.llm_manager:check_model_supports_utilities:341 - [Agent: Patient_Phoenix_Razorcrest] Mode
```

Hospital simulation example:

```
2026-09-08 06:46:34 | ERROR    | swarms.agents.llm_manager:check_model_supports_utilities:341 - [Agent: Patient_Phoenix_Razorcrest] Model 'groq/
                    Agent Name Patient_Phoenix_Razorcrest [Loop: 1/1]
It's about an 8 out of 10 right now.

It feels like a heavy pressure in the middle of my chest, and it's going into my left arm. I'm still short of
breath and sweaty. This started a little while ago today and it doesn't seem to be getting better. I have high
blood pressure and diabetes, and I take amlodipine and metformin. No known allergies.

                    Agent Name Nurse Raven Stormborn [Loop: 1/1]
Phoenix, thank you. Based on what you've told me, this is an Emergency priority.

Your triage findings are:
- Chest pain: 8/10
- Pain description: heavy pressure in the center of the chest going into the left arm
- Associated symptoms: shortness of breath and sweating
- Symptoms started: a little while ago today and not improving
- Medical history: high blood pressure, diabetes
- Medications: amlodipine, metformin
- Allergies: no known allergies

Your vital signs are:
- Blood pressure: 168/102
- Heart rate: 112 beats per minute
- Respiratory rate: 24 breaths per minute
- Temperature: 98.6°F
- Oxygen saturation: 94% on room air

Because of your chest pain, shortness of breath, sweating, pain going to the left arm, and these vital signs,
you need immediate assessment.

Next steps:
- I'm moving you for immediate emergency evaluation now.
- I want you to stop walking and stay seated or on the stretcher.
- We need a doctor to see you right away.
- You will likely have a heart monitor placed and an ECG done urgently.
- We'll continue to monitor your vital signs closely.

You're in the right place, and we're going to move quickly on this. If your pain gets worse, your breathing
becomes harder, or you feel faint, tell me immediately.

2026-09-08 06:46:40 | ERROR    | swarms.agents.llm_manager:check_model_supports_utilities:341 - [Agent: Patient_Phoenix_Razorcrest] Model 'groq/
                    Agent Name Patient_Phoenix_Razorcrest [Loop: 1/1]
Okay. Thank you.

I'm still having the chest pressure and the pain in my left arm, and I'm still short of breath. I'll stay
seated. I'm glad you're moving quickly because this really scares me.

If this could be my heart, am I going to be seen right away?
```

Hospital simulation example:

```
2026-09-08 06:46:41 | INFO     | hospital_sim.main:process_patient_queue:1036 - Triage completed for Phoenix Razorcrest,
Agent Name Dr. Zara Nightingale [Loop: 1/1]
Hello, I'm Dr. Zara Nightingale. I've reviewed your triage vitals: your blood pressure is elevated at 168/102,
your heart rate is fast at 112, and your oxygen level is a bit lower than I'd like at 94%. I want to assess you
promptly.

First, I'm starting with the basics:

- Are you having any trouble breathing right now?
- Are you having chest pain, pressure, or tightness?
- Do you feel faint, confused, or unusually weak?
- What brought you in today?

I also need a few focused details:

- When did your symptoms start?
- Where exactly is the problem or discomfort?
- Is it constant or does it come and go?
- On a scale of 0 to 10, how severe is it?
- Do you have any history of high blood pressure, heart disease, lung disease, blood clots, diabetes, or kidney
problems?
- What medications do you take regularly?
- Do you smoke, use alcohol, or any drugs?
- Any recent illness, fever, cough, travel, surgery, or leg swelling?

While we talk, I would perform an immediate exam including:

- General appearance: whether you look distressed or comfortable
- Airway: making sure you are speaking clearly and protecting your airway
- Breathing: respiratory rate, work of breathing, listening to your lungs
- Circulation: skin color, pulses, capillary refill
- Cardiac exam: heart rate and rhythm
- Neurologic screening: alertness, orientation, focal weakness
- Focused exam based on your main complaint

Based on the triage numbers alone, possibilities I would keep in mind include:

- Pain or anxiety causing elevated heart rate and blood pressure
- Heart or lung causes if you have chest pain or shortness of breath
- Infection less likely with a normal temperature, but still possible
- Dehydration
- Pulmonary issue if oxygen is truly low
- Hypertensive urgency or emergency depending on symptoms and organ involvement

Initial tests I may order depending on your symptoms and exam:

- Repeat full vital signs
- ECG
- Chest X-ray
- Bloodwork: CBC, CMP, troponin, possibly BNP
- Blood glucose
- COVID/flu testing if respiratory symptoms
- D-dimer or CT scan only if symptoms suggest a clot
- Urinalysis if indicated

Immediate treatment would depend on what you tell me and what I find on exam, but could include:

- Oxygen if your saturation remains low or you are short of breath
- Pain control if pain is driving the vital sign abnormalities
- IV access and fluids if dehydrated
- Cardiac monitoring
- Targeted treatment once the cause is clearer

Please tell me:
1. What symptoms are you having right now?
2. Do you have chest pain or shortness of breath?
3. Do you have any known medical problems or medications?

2026-09-08 06:46:50 | ERROR    | swarms.agents.llm_manager:check_model_supports_utilities:341 - [Agent: Patient_Phoenix_R
Agent Name Patient_Phoenix_Razorcrest [Loop: 1/1]
Hi, Doctor.

Yes, I'm having chest pain right now, and I'm short of breath. It feels like a heavy pressure in the middle of
my chest, and it's going down my left arm. I'm also sweating a lot.

It started a little while ago today and it's been pretty constant since it started. Right now I'd say it's
about an 8 out of 10. I don't feel confused, but I do feel weak and scared. I don't think I'm going to pass
out, but I definitely feel bad.

I have high blood pressure and diabetes. I take amlodipine and metformin regularly. I don't have any known
allergies.

I don't have lung disease, blood clots, or kidney problems that I know of. No fever or cough. No recent surgery
or travel. I haven't noticed any leg swelling.
```

Hospital simulation example:

Agent Name Dr. Zara Nightingale [Loop: 1/1]

Phoenix, I'm very concerned this is an acute coronary syndrome, meaning a heart-related emergency, and I have to treat it as a possible heart attack until proven otherwise.

Final working diagnosis:
- Acute coronary syndrome, highly concerning for myocardial infarction
- Chest pain with classic cardiac features: heavy central pressure, radiation to left arm, shortness of breath, diaphoresis
- Abnormal vital signs: hypertension, tachycardia, mild hypoxemia
- Cardiac risk factors: hypertension and diabetes

Other diagnoses I would still consider, but they are less likely based on what you've told me:
- Unstable angina
- Aortic syndrome
- Pulmonary embolism
- Less likely musculoskeletal pain, reflux, or anxiety alone

My focused examination findings, based on the information available, are concerning for distress but with airway intact:
- Airway: speaking, so airway is patent
- Breathing: short of breath, oxygen saturation 94%, increased respiratory effort possible
- Circulation: elevated blood pressure 168/102, heart rate 112, ongoing chest pain and diaphoresis
- Neurologic: awake, oriented, no confusion reported
- General: appears acutely ill/anxious due to pain and possible cardiac ischemia

Immediate treatment plan:
- Place on continuous cardiac monitor
- Obtain a 12-lead ECG immediately
- Establish IV access
- Give aspirin now, if not already given and no contraindication
- Give oxygen only if saturation drops further or respiratory distress increases; monitor closely
- Check blood glucose given diabetes
- Draw urgent labs:
  - Troponin serially
  - CBC
  - CMP/BMP
  - Magnesium
  - Coagulation studies as needed
- Portable chest X-ray
- Repeat vital signs frequently

Symptom-directed emergency medications I would consider:
- Nitroglycerin if blood pressure remains adequate and there is no contraindication such as recent erectile dysfunction medication use or suspicion for right-sided infarct
- Stronger pain control only as needed after initial cardiac assessment
- If ECG confirms STEMI, activate the cath lab immediately for emergent reperfusion
- If NSTEMI/unstable angina is suspected, start guideline-directed ACS treatment and admit for cardiology management

Disposition:
- This is not safe for discharge
- You need immediate emergency department management and hospital admission
- Cardiology consultation is indicated urgently

What I would document in the EHR:
- 45-year-old patient with HTN and diabetes presenting with constant 8/10 substernal pressure radiating to left arm, associated dyspnea and diaphoresis, onset today
- Vitals significant for BP 168/102, HR 112, RR 24, SpO2 94%, afebrile

Hospital simulation example:

Only partial output is shown here. After having the complete output, the following information is diplayed.

```
2026-09-08 06:48:08 | INFO     | hospital_sim.main:add_patient_record:288 - Adding patient record to EHR: Sage Whisperwind (ID: ae86ac7e-9d9a-45ed-b1a6-004b14db4144) - Doctor: Dr. Zara Nightingale
2026-09-08 06:48:11 | INFO     | hospital_sim.main:process_patient_queue:1103 - Patient Sage Whisperwind treated by Dr. Zara Nightingale - Diagnosis: Sage, my final diagnosis is: - Time: 1.9 minutes
✅ Patient Sage Whisperwind treated by Dr. Zara Nightingale
   Diagnosis: Sage, my final diagnosis is:
   Treatment Time: 1.9 minutes

💾 Colab Emergency Medical Center - Final Report2026-09-08 06:48:29 | INFO     | hospital_sim.main:run_simulation:1722 - Simulation ended

==================================================
Simulation Duration: 0:02:15.046577
Total Patients: 3
Patients Treated: 3
Average Wait Time: 1.4 minutes
Total Revenue: $450.00
Total Costs: $300.00
Net Profit: $150.00
Patient Satisfaction: 97.7%

Staff Performance:
  Dr. Zara Nightingale: 3 patients, Avg time: 1.4 min

EHR System: ChromaDB with collection 'hospital_ehr'
```

Thank You For Listening !

Questions, Comments & Suggestions are Welcome